Boss Challenge — Paper 28 (Class 7)

Heat | Reproduction in Plants | Integers | Symmetry

One correct option each. Marking: +4 correct, -1 wrong, 0 blank. Write A/B/C/D on a separate sheet.

1. The degree of hotness or coldness of a body, telling us how hot or cold it is, is called its:

- **(A)** length
- **(B)** volume
- **(C)** temperature
- **(D)** weight

2. The process by which living organisms produce new individuals of their own kind is called:

- **(A)** respiration
- **(B)** reproduction
- **(C)** germination
- **(D)** digestion

3. The collection of numbers that includes the positive whole numbers, zero and the negative whole numbers - such as …, -2, -1, 0, 1, 2, … - is called the:

- **(A)** decimals
- **(B)** fractions
- **(C)** only positive numbers
- **(D)** integers

4. A figure that can be folded along a line so that its two halves fall exactly on top of each other is said to have:

- **(A)** no symmetry
- **(B)** curved symmetry
- **(C)** rotational symmetry only
- **(D)** line symmetry

5. The instrument that is used to measure the temperature of a body accurately is the:

- **(A)** speedometer
- **(B)** barometer
- **(C)** thermometer
- **(D)** ammeter

6. When a new plant is produced from a single parent, without seeds and without the joining of two cells, the reproduction is called:

- **(A)** pollination
- **(B)** asexual reproduction
- **(C)** germination
- **(D)** sexual reproduction

7. Among all the integers, the one number that is neither positive nor negative is:

- **(A)** 1
- **(B)** 10
- **(C)** -1
- **(D)** 0

8. The line along which a figure is folded so that the two halves match exactly is called its:

- **(A)** line of symmetry
- **(B)** number line
- **(C)** base line
- **(D)** diagonal only

9. The special thermometer kept by a doctor to measure the temperature of the human body is the:

- **(A)** digital clock
- **(B)** laboratory thermometer
- **(C)** maximum-minimum thermometer
- **(D)** clinical thermometer

10. Reproduction that involves two parents and the joining of a male and a female cell, usually giving seeds, is called:

- **(A)** sexual reproduction
- **(B)** vegetative propagation
- **(C)** asexual reproduction
- **(D)** budding

11. On a number line drawn in the usual way, the integers that lie to the left of zero are all:

- **(A)** negative
- **(B)** positive
- **(C)** fractions
- **(D)** zero

12. The number of lines of symmetry that a square has is:

- **(A)** 1
- **(B)** 4
- **(C)** 0
- **(D)** 2

13. The temperature of a healthy human body, the value a doctor expects to read, is about:

- **(A)** 100°C
- **(B)** 0°C
- **(C)** 37°C
- **(D)** 50°C

14. In yeast, a small bulge grows out from the parent cell, enlarges and finally separates as a new yeast. This method is called:

- **(A)** fragmentation
- **(B)** pollination
- **(C)** budding
- **(D)** fertilisation

15. When you add two negative integers together, such as (-3) + (-4), the answer is always:

- **(A)** either positive or negative
- **(B)** zero
- **(C)** positive
- **(D)** negative

16. The number of lines of symmetry that a rectangle (which is not a square) has is:

- **(A)** 2
- **(B)** 1
- **(C)** 4
- **(D)** 0

17. A clinical thermometer has a small bend, called a kink, in its tube. The kink stops the mercury from:

- **(A)** rising too fast
- **(B)** changing colour
- **(C)** turning into gas
- **(D)** slipping back on its own

18. New potato plants can sprout from the small buds, called 'eyes', on a potato. Growing a new plant from such a plant part is called:

- **(A)** vegetative propagation
- **(B)** pollination
- **(C)** seed dispersal
- **(D)** fertilisation

19. The value of (-5) + 8 is:

- **(A)** 13
- **(B)** -13
- **(C)** -3
- **(D)** 3

20. The number of lines of symmetry that an equilateral triangle has is:

- **(A)** 2
- **(B)** 1
- **(C)** 0
- **(D)** 3

21. For measuring the temperature of things other than the body, such as hot water in an experiment, we use a:

- **(A)** compass
- **(B)** weighing scale
- **(C)** laboratory thermometer
- **(D)** clinical thermometer

22. The leaves of the Bryophyllum plant carry tiny buds along their edges, and each bud can grow into a new plant. This shows vegetative propagation by:

- **(A)** seeds
- **(B)** flowers
- **(C)** leaves
- **(D)** roots

23. Adding the two negatives in (-6) + (-9) gives a result of:

- **(A)** -3
- **(B)** 15
- **(C)** 3
- **(D)** -15

24. The number of lines of symmetry that a circle has is:

- **(A)** infinitely many
- **(B)** 4
- **(C)** 0
- **(D)** only 1

25. When a hot cup of tea is left on a table, heat flows on its own from the hotter tea to the cooler surroundings, that is, always from a:

- **(A)** colder body to a hotter body
- **(B)** hotter body to a colder body
- **(C)** larger body to a smaller body
- **(D)** heavier body to a lighter body

26. Mosses and ferns often reproduce by tiny, light cells that are scattered by the wind and grow into new plants. These cells are called:

- **(A)** roots
- **(B)** spores
- **(C)** seeds
- **(D)** buds

27. The value of 7 - 10 is:

- **(A)** 3
- **(B)** 17
- **(C)** -17
- **(D)** -3

28. The line of symmetry of the capital letter A is:

- **(A)** a vertical line down the middle
- **(B)** a horizontal line across the middle
- **(C)** no line at all
- **(D)** a diagonal line

29. The way heat travels through a solid metal rod from the hot end to the cold end, without the metal itself moving, is called:

- **(A)** convection
- **(B)** conduction
- **(C)** reflection
- **(D)** radiation

30. The male reproductive part of a flower, which makes the pollen, is called the:

- **(A)** petal
- **(B)** stamen
- **(C)** sepal
- **(D)** pistil

31. Working out (-4) - (-9), where subtracting a negative turns into adding, gives:

- **(A)** -13
- **(B)** 5
- **(C)** -5
- **(D)** 13

32. The number of lines of symmetry that the capital letter H has is:

- **(A)** 1
- **(B)** 0
- **(C)** 2
- **(D)** 4

33. Materials such as iron, copper and aluminium, which let heat pass through them easily, are called:

- **(A)** insulators
- **(B)** liquids
- **(C)** conductors
- **(D)** gases

34. The female reproductive part of a flower, which contains the ovary, is called the:

- **(A)** sepal
- **(B)** petal
- **(C)** pistil
- **(D)** stamen

35. The product (-3) x 4 works out to:

- **(A)** -12
- **(B)** 12
- **(C)** -7
- **(D)** 1

36. The number of lines of symmetry that an isosceles triangle (two equal sides) has is:

- **(A)** 0
- **(B)** 2
- **(C)** 3
- **(D)** 1

37. Materials such as wood, plastic and cloth, which do not let heat pass through them easily, are called:

- **(A)** insulators
- **(B)** conductors
- **(C)** metals
- **(D)** fuels

38. The sticky top of the pistil, on which pollen grains land and are caught, is called the:

- **(A)** anther
- **(B)** ovary
- **(C)** stigma
- **(D)** filament

39. The value of (-5) x (-6) is:

- **(A)** 30
- **(B)** -30
- **(C)** 11
- **(D)** -11

40. The number of lines of symmetry that a scalene triangle (all sides different) has is:

- **(A)** 1
- **(B)** 3
- **(C)** 0
- **(D)** infinitely many

41. In water and air, heat travels by the actual movement of the heated portions rising and the cooler ones sinking. This way of heat transfer is called:

- **(A)** radiation
- **(B)** convection
- **(C)** freezing
- **(D)** conduction

42. Pollen grains are made and stored in a part at the top of the stamen called the:

- **(A)** ovule
- **(B)** anther
- **(C)** sepal
- **(D)** stigma

43. When two negative integers are multiplied together, the product is always:

- **(A)** negative
- **(B)** sometimes negative
- **(C)** positive
- **(D)** zero

44. When a figure is turned about a fixed point and still looks exactly the same before a full turn is complete, it is said to have:

- **(A)** no symmetry
- **(B)** rotational symmetry
- **(C)** translation symmetry
- **(D)** line symmetry

45. Heat from the Sun reaches the Earth across the empty space where there is no air or material. This way of heat travel is called:

- **(A)** evaporation
- **(B)** radiation
- **(C)** conduction
- **(D)** convection

46. The transfer of pollen grains from the anther of a flower to the stigma is called:

- **(A)** fertilisation
- **(B)** pollination
- **(C)** respiration
- **(D)** germination

47. Dividing (-20) / 5 gives a quotient of:

- **(A)** -15
- **(B)** 4
- **(C)** -4
- **(D)** -100

48. The fixed point about which a figure is rotated when we test its rotational symmetry is called the:

- **(A)** midpoint of a side
- **(B)** centre of rotation
- **(C)** line of symmetry
- **(D)** vertex only

49. On a hot, sunny day we feel cooler in light-coloured clothes because light or white surfaces:

- **(A)** reflect most of the heat
- **(B)** turn heat into light
- **(C)** absorb most of the heat
- **(D)** make their own heat

50. When pollen from a flower is transferred to the stigma of the same flower or another flower on the same plant, it is called:

- **(A)** dispersal
- **(B)** fertilisation
- **(C)** self-pollination
- **(D)** cross-pollination

51. The value of (-36) / (-6) is:

- **(A)** -6
- **(B)** 6
- **(C)** 30
- **(D)** -42

52. As a square is turned through one full circle about its centre, the number of times it looks exactly the same (its order of rotational symmetry) is:

- **(A)** 2
- **(B)** 1
- **(C)** 8
- **(D)** 4

53. In winter we often prefer dark-coloured clothes because dark surfaces:

- **(A)** absorb more heat
- **(B)** cool the body down
- **(C)** reflect more heat
- **(D)** block all heat

54. When pollen from one plant's flower is carried to the stigma of a flower on a different plant of the same kind, it is called:

- **(A)** self-pollination
- **(B)** germination
- **(C)** fertilisation
- **(D)** cross-pollination

55. The additive inverse (the number that you add to it to get zero) of the integer 7 is:

- **(A)** 1/7
- **(B)** -7
- **(C)** 0
- **(D)** 7

56. The smallest angle through which a square must be turned about its centre to look exactly the same again is:

- **(A)** 360°
- **(B)** 45°
- **(C)** 90°
- **(D)** 180°

57. During the day the land heats up faster than the sea, so a cool wind blows from the sea towards the land. This is called the:

- **(A)** cyclone
- **(B)** monsoon wind
- **(C)** sea breeze
- **(D)** land breeze

58. Flowers depend on outside helpers to carry their pollen about. The common natural agents of pollination are:

- **(A)** only sunlight
- **(B)** only earthworms
- **(C)** only electricity
- **(D)** wind, water and insects

59. On the number line, the integer that comes just to the right of -1 (one step larger) is:

- **(A)** 1
- **(B)** -2
- **(C)** -1
- **(D)** 0

60. A figure that looks exactly the same after a half turn (a turn of 180°) about its centre has rotational symmetry of order:

- **(A)** 1
- **(B)** 3
- **(C)** 4
- **(D)** 2

61. At night the land cools faster than the sea, so a wind blows from the land towards the sea. This is called the:

- **(A)** land breeze
- **(B)** storm
- **(C)** whirlwind
- **(D)** sea breeze

62. Once fertilisation is over, the whole ovary of the flower swells and ripens to become the:

- **(A)** root
- **(B)** stem
- **(C)** fruit
- **(D)** leaf

63. Among the integers -3, -7, 2 and 0, the greatest (largest) one is:

- **(A)** -7
- **(B)** -3
- **(C)** 2
- **(D)** 0

64. A regular five-petalled flower looks just like a regular pentagon shape. The number of lines of symmetry it has is:

- **(A)** 1
- **(B)** 5
- **(C)** 10
- **(D)** 2

65. Woollen clothes keep us warm in winter mainly because the wool fibres trap a layer of:

- **(A)** air, which is a poor conductor of heat
- **(B)** metal, which conducts heat
- **(C)** heat made by the wool itself
- **(D)** water, which carries heat away

66. After fertilisation, each ovule inside the ovary develops into a:

- **(A)** seed
- **(B)** petal
- **(C)** flower
- **(D)** root

67. At dawn a hill station reads -5°C. By noon the temperature has risen by 8°C. Treating the rise as adding +8, the noon temperature is:

- **(A)** 13°C
- **(B)** 3°C
- **(C)** -13°C
- **(D)** -3°C

68. When a simple leaf is folded along its midrib (the central vein), the two halves match. For that leaf the midrib acts as a:

- **(A)** line of symmetry
- **(B)** axis of rotation
- **(C)** diagonal of a square
- **(D)** centre of rotation

69. Wearing two thin blankets can keep you warmer than one thick blanket because the gap between them traps:

- **(A)** a layer of air
- **(B)** sunlight
- **(C)** cold water
- **(D)** extra heat from outside

70. The actual joining, or fusion, of a male cell from the pollen with the female cell (egg) in the ovule is called:

- **(A)** pollination
- **(B)** germination
- **(C)** budding
- **(D)** fertilisation

71. One evening a town is at 4°C. Overnight the temperature falls by 10°C. Treating the fall as subtracting 10, the morning temperature is:

- **(A)** -6°C
- **(B)** -14°C
- **(C)** 6°C
- **(D)** 14°C

72. The two wings of a butterfly are mirror images of each other. The line of symmetry of the butterfly passes:

- **(A)** through one wing only
- **(B)** around the edge of a wing
- **(C)** down the middle of its body
- **(D)** across the tips of the wings

73. The silvery liquid that rises and falls inside many common thermometers as the temperature changes is:

- **(A)** water
- **(B)** mercury
- **(C)** oil
- **(D)** milk

74. A flower that has both stamens (the male part) and a pistil (the female part) in the same flower is called a:

- **(A)** seedless flower
- **(B)** bisexual flower
- **(C)** unisexual flower
- **(D)** petal-less flower

75. A deep freezer is kept at -12°C. When its door is left open, the inside warms up by 5°C. Adding +5, the new temperature is:

- **(A)** -5°C
- **(B)** 7°C
- **(C)** -17°C
- **(D)** -7°C

76. A figure that looks the same only after a complete full turn of 360° (and at no smaller turn) is said to have rotational symmetry of order:

- **(A)** 1
- **(B)** 2
- **(C)** 4
- **(D)** 0

77. Before using a clinical thermometer, a doctor gives it a few sharp jerks. This is done to:

- **(A)** clean the glass outside
- **(B)** make the mercury boil
- **(C)** warm the thermometer up
- **(D)** bring the mercury down below normal

78. A flower that has only stamens or only a pistil, but not both, is called a:

- **(A)** double flower
- **(B)** unisexual flower
- **(C)** complete flower
- **(D)** bisexual flower

79. When any integer is multiplied by (-1), the result you obtain is:

- **(A)** always a positive number
- **(B)** zero every time
- **(C)** the same number with its sign changed
- **(D)** the same number unchanged

80. The image of a shape seen in a mirror placed along a line, flipped to the other side of that line, is called its:

- **(A)** rotation
- **(B)** translation
- **(C)** reflection
- **(D)** enlargement

81. When a steel spoon is left standing in a cup of hot tea, its top end soon becomes warm too. This happens by:

- **(A)** convection
- **(B)** radiation
- **(C)** conduction
- **(D)** melting

82. Some seeds, such as those of the drumstick and maple, have wing-like flaps that help them float away on moving air. These seeds are dispersed by:

- **(A)** water
- **(B)** explosion
- **(C)** animals
- **(D)** wind

83. The value of 0 x (-8) is:

- **(A)** -8
- **(B)** 8
- **(C)** -80
- **(D)** 0

84. The number of lines of symmetry that a regular hexagon (six equal sides) has is:

- **(A)** 12
- **(B)** 3
- **(C)** 6
- **(D)** 4

85. The handles of cooking pans are often made of plastic or wood rather than metal because these materials are good:

- **(A)** conductors
- **(B)** insulators
- **(C)** reflectors
- **(D)** fuels

86. Some seeds, such as those of Xanthium, have tiny hooks or spines that cling to the fur of passing animals. These seeds are dispersed by:

- **(A)** animals
- **(B)** wind
- **(C)** the plant itself
- **(D)** water

87. The value of (-15) + 15 is:

- **(A)** 1
- **(B)** 30
- **(C)** 0
- **(D)** -30

88. The capital letter S looks the same after a half turn about its centre but cannot be folded into matching halves. So the letter S has:

- **(A)** rotational symmetry but no line symmetry
- **(B)** neither kind of symmetry
- **(C)** line symmetry but no rotational symmetry
- **(D)** both line and rotational symmetry

89. A clinical thermometer's scale usually runs only over a narrow range, from about:

- **(A)** 35°C to 42°C
- **(B)** 0°C to 100°C
- **(C)** 100°C to 200°C
- **(D)** -10°C to 50°C

90. A coconut can float on the sea for a long way before sprouting on a distant shore. Such floating seeds are dispersed by:

- **(A)** magnetism
- **(B)** wind
- **(C)** insects
- **(D)** water

91. The number that must be added to -8 to give a total of 0 is:

- **(A)** 0
- **(B)** 8
- **(C)** -8
- **(D)** 16

92. A rangoli pattern whose left half is the exact mirror image of its right half is showing:

- **(A)** translation only
- **(B)** line symmetry
- **(C)** rotational symmetry only
- **(D)** no symmetry

93. When a metal ball is heated strongly, it gets slightly bigger. This increase in size on heating is called:

- **(A)** thermal expansion
- **(B)** evaporation
- **(C)** conduction
- **(D)** freezing

94. Seed dispersal - the scattering of seeds away from the parent plant - is important mainly because it prevents the new plants from:

- **(A)** growing roots at all
- **(B)** being eaten by anything
- **(C)** ever making flowers
- **(D)** overcrowding and competing for the same space, water and light

95. On a cold day a kitchen is at 6°C while the freezer is at -4°C. The difference between these two temperatures (kitchen minus freezer) is:

- **(A)** 10°C
- **(B)** 24°C
- **(C)** 2°C
- **(D)** -10°C

96. An equilateral triangle is turned about its centre through one full circle. The number of times it looks exactly the same (its order of rotational symmetry) is:

- **(A)** 2
- **(B)** 6
- **(C)** 3
- **(D)** 1

97. In short, heat is best described as a form of:

- **(A)** energy that flows because of a temperature difference
- **(B)** light made only by the Sun
- **(C)** a liquid stored inside hot objects
- **(D)** matter that you can weigh

98. A money-plant grown by placing a cut piece of its stem in water, where it forms roots and grows, is an example of:

- **(A)** pollination
- **(B)** fertilisation
- **(C)** seed germination
- **(D)** vegetative propagation

99. The property shown when 5 + (-5) = 0 - that every integer has an opposite which adds to give zero - is the:

- **(A)** closure property
- **(B)** multiplicative inverse property
- **(C)** distributive property
- **(D)** additive inverse property

100. The number of lines of symmetry that a rhombus (a slanted four-sided shape with all sides equal) has is:

- **(A)** 1
- **(B)** 2
- **(C)** 0
- **(D)** 4